TUTORIAL 13

Gram--Schmidt and Orthogonal Diagonalization

MOTIVATION AND GUIDING IDEA

Gram--Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

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DEFINITION 13.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram--Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j / \|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w \rangle = \langle v, T^* w \rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad RR^T = R^T R = I,$$

but $R^2 = -I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram--Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

MOTIVATION AND GUIDING IDEA

We construct orthonormal bases by systematically subtracting projections and compute projections in polynomial spaces. We study the conditions for orthogonal diagonalization over the reals and unitary diagonalization over the complex numbers, and use an orthonormal eigenbasis to prove normality.

13.1 Review and Notation

RECALL AND TOOLS

Throughout this tutorial, the inner product space is over $\mathbb{R}$ or $\mathbb{C}$; in the complex case, our convention is:

1. $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0 \iff x = 0$.
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
3. Linearity in the first argument: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.

It follows that the second argument is conjugate linear: $\langle x, ay + bz \rangle = \overline{a}\langle x, y \rangle + \overline{b}\langle x, z \rangle$. This is the same convention as in the preceding tutorial; over $\mathbb{R}$, the two linearity conventions coincide.

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DEFINITION 13.2

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. Vectors $u, v \in V$ are orthogonal if $\langle u, v\rangle = 0$.
2. A list $(e_1, \ldots, e_k)$ is orthonormal if $\langle e_i, e_j\rangle = \delta_{ij}$ for all i, j.
3. For each $x \in V$, define the norm $\|x\| := \sqrt{\langle x, x\rangle}$.

RECALL AND TOOLS

Orthogonal projection onto a line: if $u \neq 0$, the vector

$$\text{proj}_{\text{span}\{u\}}(v) := \frac{\langle v, u\rangle}{\langle u, u\rangle}\, u$$

is the unique vector of the form αu for which $v - \alpha u \perp u$.

13.2 The Gram--Schmidt Algorithm

We present Gram--Schmidt with immediate normalization at each step, constructing an orthonormal list directly.

Let $v_1, \ldots, v_k \in V$ be linearly independent. Define recursively:

1. Set $w_1 := v_1$, and then $e_1 := \dfrac{w_1}{\|w_1\|}$.
2. For $j = 2, \ldots, k$, first define
$$w_j := v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle\, e_i,$$
and then normalize:
$$e_j := \frac{w_j}{\|w_j\|}.$$

The main conclusions are:

1. $(e_1, \ldots, e_k)$ is orthonormal.
2. For every m, $\text{span}\{v_1, \ldots, v_m\} = \text{span}\{e_1, \ldots, e_m\}$.

Formal justification: assume inductively that $e_1, \ldots, e_{j-1}$ are orthonormal. For every $r < j$,

$$\begin{aligned}\langle w_j, e_r\rangle &= \left\langle v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle e_i,\ e_r \right\rangle \\ &= \langle v_j, e_r\rangle - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle \langle e_i, e_r\rangle \\ &= \langle v_j, e_r\rangle - \langle v_j, e_r\rangle = 0.\end{aligned}$$

If $w_j = 0$, then v_j belongs to the span of $e_1, \ldots, e_{j-1}$, which by induction equals the span of $v_1, \ldots, v_{j-1}$. This contradicts the linear independence of the input. Thus $w_j \neq 0$, normalization is permitted, and $\|e_j\| = 1$. Also, $w_j \in \text{span}\{v_1, \ldots, v_j\}$, and the subtraction formula implies $v_j \in \text{span}\{e_1, \ldots, e_{j-1}, w_j\}$. Hence the corresponding prefix spans agree at every step.

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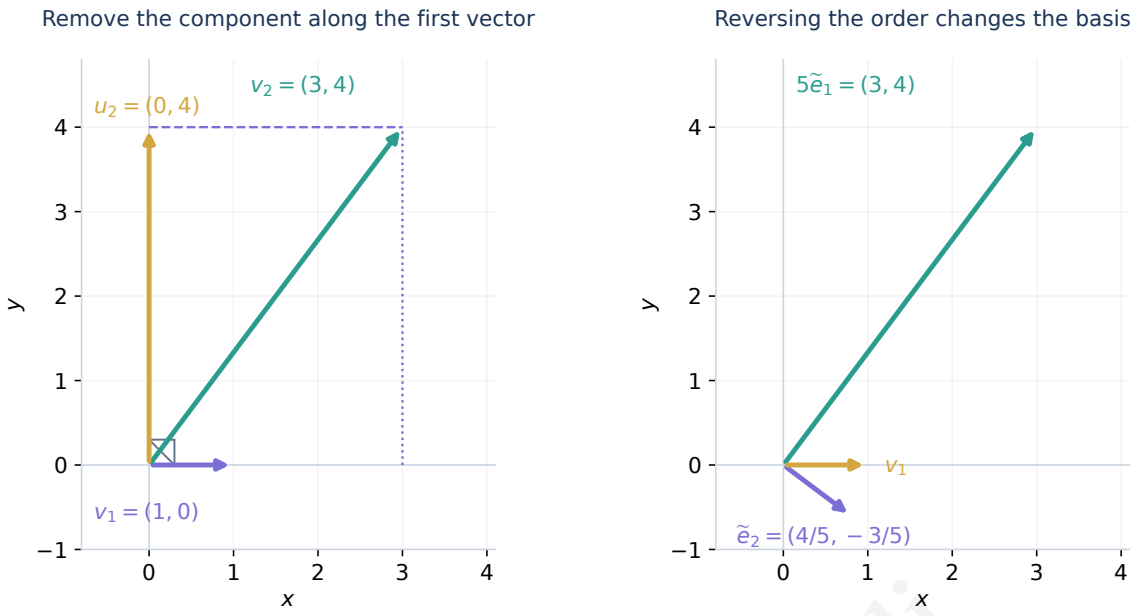


Figure 13.1. At each step, subtract from the next vector its components in the directions already constructed. The residual is orthogonal to the preceding directions; normalization produces the next unit vector.

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EXERCISE 13.1

Apply Gram--Schmidt, normalizing at every step, to the vectors

$$v_1 = (3, 4, 0)^T, \quad v_2 = (1, 3, 0)^T, \quad v_3 = (2, 1, 5)^T$$

in $\mathbb{R}^3$ with the standard inner product. Explicitly verify that the output is orthonormal.

WORKED SOLUTION

The first norm is $\|v_1\| = \sqrt{9 + 16} = 5$, so $e_1 = (3/5, 4/5, 0)^T$. Compute the projection of v_2:

$$\langle v_2, e_1 \rangle = \frac{3}{5} + \frac{12}{5} = 3,$$

$$w_2 = v_2 - 3e_1 = (1 - 9/5, 3 - 12/5, 0)^T = (-4/5, 3/5, 0)^T.$$

Since $\|w_2\|^2 = 16/25 + 9/25 = 1$, we have $e_2 = w_2$. The two coefficients for v_3 are

$$\langle v_3, e_1 \rangle = 6/5 + 4/5 = 2, \qquad \langle v_3, e_2 \rangle = -8/5 + 3/5 = -1.$$

Thus

$$w_3 = v_3 - 2e_1 - (-1)e_2 = (2 - 6/5 - 4/5, 1 - 8/5 + 3/5, 5)^T = (0, 0, 5)^T.$$

Its norm is 5, so $e_3 = (0, 0, 1)^T$. The output is

$$e_1 = \tfrac{1}{5}(3, 4, 0)^T, \qquad e_2 = \tfrac{1}{5}(-4, 3, 0)^T, \qquad e_3 = (0, 0, 1)^T.$$

The first two norms are $\sqrt{25/25} = 1$, and the third is 1. The inner product $\langle e_1, e_2 \rangle = (-12 + 12)/25 = 0$, and both inner products with e_3 vanish. We have obtained an orthonormal basis with rational coordinates only.

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EXERCISE 13.2

In $\mathbb{C}^2$ with $\langle x, y\rangle = x_1\overline{y_1} + x_2\overline{y_2}$, let $v_1 = (3, 4i)^T$ and $v_2 = (1, i)^T$. Apply Gram--Schmidt with immediate normalization, taking care to conjugate the correct argument in the projection coefficient.

WORKED SOLUTION

First, $\|v_1\|^2 = 9 + |4i|^2 = 25$, so $e_1 = (3, 4i)^T/5$. With the convention of linearity in the first argument,

$$\langle v_2, e_1\rangle = 1 \cdot \frac{3}{5} + i \cdot \frac{\overline{4i}}{5} = \frac{3}{5} + i\frac{-4i}{5} = \frac{7}{5}.$$

Subtract the projection:

$$w_2 = v_2 - \frac{7}{5}e_1 = \left(1 - \frac{21}{25}, i - \frac{28i}{25}\right)^T = \frac{1}{25}(4, -3i)^T.$$

Thus $\|w_2\|^2 = (16 + 9)/625 = 1/25$, so $\|w_2\| = 1/5$, and $e_2 = (4, -3i)^T/5$. Check:

$$\|e_2\|^2 = \frac{16 + 9}{25} = 1, \qquad \langle e_1, e_2\rangle = \frac{3 \cdot 4 + (4i)(3i)}{25} = 0.$$

The output (e_1, e_2) is orthonormal. The product $i\overline{4i} = -4i^2 = 4$ illustrates why conjugation is essential.

EXERCISE 13.3

On $\mathbb{R}[t]_{\le 2}$, define the inner product

$$\langle p, q\rangle := \int_0^1 p(t)\, q(t)\, dt.$$

Apply Gram--Schmidt to $(v_1, v_2, v_3) = (1, t, t^2)$ to obtain an orthonormal list (e_1, e_2, e_3), normalizing immediately at every step.

WORKED SOLUTION

1. $$\|1\|^2 = \int_0^1 1\, dt = 1 \quad \Rightarrow \quad e_1 = 1.$$

2. Compute
$$\langle t, e_1\rangle = \int_0^1 1 \cdot t\, dt = \frac{1}{2}, \qquad w_2 = t - \langle t, e_1\rangle e_1 = t - \frac{1}{2}.$$
Norm:
$$\|w_2\|^2 = \int_0^1 \left(t - \frac{1}{2}\right)^2 dt = \int_0^1 \left(t^2 - t + \frac{1}{4}\right) dt = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12},$$
and hence
$$e_2 = \frac{w_2}{\|w_2\|} = \frac{t - \frac{1}{2}}{\sqrt{1/12}} = 2\sqrt{3}\left(t - \frac{1}{2}\right).$$

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EXERCISE 13.3 CONTINUED

3. First compute

$$\langle t^2, e_1\rangle = \int_0^1 t^2 dt = \frac{1}{3}.$$

and

$$\langle t^2, e_2\rangle = \int_0^1 2\sqrt{3}\left(t-\frac{1}{2}\right)t^2 dt = 2\sqrt{3}\int_0^1 \left(t^3-\frac{1}{2}t^2\right)dt = 2\sqrt{3}\left(\frac{1}{4}-\frac{1}{6}\right) = \frac{\sqrt{3}}{6}.$$

Thus

$$\begin{aligned} w_3 &= t^2 - \langle t^2, e_1\rangle e_1 - \langle t^2, e_2\rangle e_2 \\ &= t^2 - \frac{1}{3} - \frac{\sqrt{3}}{6}\cdot 2\sqrt{3}\left(t-\frac{1}{2}\right) \\ &= t^2 - \frac{1}{3} - \left(t-\frac{1}{2}\right) \\ &= t^2 - t + \frac{1}{6}. \end{aligned}$$

Norm:

$$\begin{aligned} \|w_3\|^2 &= \int_0^1 \left(t^2 - t + \frac{1}{6}\right)^2 dt \\ &= \int_0^1 \left(t^4 - 2t^3 + \frac{4}{3}t^2 - \frac{1}{3}t + \frac{1}{36}\right) dt \\ &= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}, \end{aligned}$$

Therefore

$$\|w_3\| = \sqrt{\frac{1}{180}} = \frac{1}{6\sqrt{5}}.$$

and hence

$$e_3 = \frac{w_3}{\|w_3\|} = 6\sqrt{5}\left(t^2 - t + \frac{1}{6}\right).$$

The resulting basis is:

$$e_1 = 1, \qquad e_2 = 2\sqrt{3}\left(t-\frac{1}{2}\right), \qquad e_3 = 6\sqrt{5}\left(t^2 - t + \frac{1}{6}\right).$$

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EXERCISE 13.4

In $\mathbb{R}^2$, compare the Gram--Schmidt outputs for the two orders (v_1, v_2) and (v_2, v_1), where $v_1 = (1,0)^T$ and $v_2 = (3,4)^T$. Explain what remains the same despite the different outputs.

WORKED SOLUTION

For the first order, $e_1 = (1,0)^T$. The projection coefficient is $\langle v_2, e_1\rangle = 3$, so

$$w_2 = (3,4)^T - 3(1,0)^T = (0,4)^T, \qquad e_2 = (0,1)^T.$$

For the reversed order, start with $\widetilde{e}_1 = (3,4)^T/5$. The projection coefficient of v_1 onto it is $3/5$, so

$$\widetilde{w}_2 = (1,0)^T - \frac{3}{5}\frac{1}{5}(3,4)^T = \left(\frac{16}{25}, -\frac{12}{25}\right)^T.$$

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EXERCISE 13.4 CONTINUED

Its norm is $\sqrt{256+144}/25 = 20/25 = 4/5$. Normalization gives

$$\widetilde{e}_2 = (4/5, -3/5)^T.$$

Thus the first order gives the standard basis, whereas the second gives $((3/5, 4/5)^T, (4/5, -3/5)^T)$. Both are orthonormal and span the same plane. At every step, the span of the input prefix is preserved, but changing the input order changes the first direction and can therefore change the entire output.

EXERCISE 13.5

Let $V = \mathbb{C}_3[x]$ be the space of complex polynomials of degree at most 3. For $f, g \in V$, define

$$\langle f, g\rangle := f(-1)\,\overline{g(-1)} + f(0)\,\overline{g(0)} + f(1)\,\overline{g(1)} + f(2)\,\overline{g(2)}.$$

1. Prove that this is an inner product on V.
2. Let $W = \mathbb{C}_2[x] \subseteq V$. Find an orthonormal basis of W.
3. Find $W^{\perp}$.

WORKED SOLUTION

1. Verify the axioms.

   (a) Positive definiteness: for every $f \in V$,

   $$\langle f, f\rangle = \sum_{t\in\{-1,0,1,2\}} f(t)\,\overline{f(t)} = \sum_{t\in\{-1,0,1,2\}} |f(t)|^2 \geq 0.$$

   If $\langle f, f\rangle = 0$, then $f(-1) = f(0) = f(1) = f(2) = 0$. A nonzero polynomial of degree at most 3 has at most 3 distinct roots, so necessarily $f \equiv 0$.

   (b) Conjugate symmetry: for every $f, g \in V$,

   $$\overline{\langle g, f\rangle} = \overline{\sum_t g(t)\,\overline{f(t)}} = \sum_t f(t)\,\overline{g(t)} = \langle f, g\rangle.$$

   (c) Linearity in the first argument: for all $f, g, h \in V$ and all $\alpha, \beta \in \mathbb{C}$,

   $$\begin{aligned}\langle \alpha f + \beta g, h\rangle &= \sum_t \big(\alpha f(t) + \beta g(t)\big)\overline{h(t)} \\ &= \alpha \sum_t f(t)\overline{h(t)} + \beta \sum_t g(t)\overline{h(t)} \\ &= \alpha\langle f, h\rangle + \beta\langle g, h\rangle.\end{aligned}$$

   Thus this is an inner product on V.

2. Apply Gram--Schmidt to $(1, x, x^2)$ and normalize at every step. Write $v_1 = 1$, $v_2 = x$, $v_3 = x^2$.

   First,

   $$\|v_1\|^2 = \langle 1, 1\rangle = 1 + 1 + 1 + 1 = 4, \qquad e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2}.$$

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EXERCISE 13.5 CONTINUED

Next,

$$\langle v_2, e_1\rangle = \left\langle x, \frac{1}{2}\right\rangle = \frac{1}{2}\big((-1)+0+1+2\big) = 1, \qquad w_2 = v_2 - \langle v_2, e_1\rangle e_1 = x - \frac{1}{2}.$$

Compute the norm:

$$\begin{aligned}\|w_2\|^2 &= \left(-\frac{3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 \\ &= \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} = 5.\end{aligned}$$

Therefore

$$e_2 = \frac{w_2}{\|w_2\|} = \frac{x - \frac{1}{2}}{\sqrt{5}}.$$

Next,

$$\langle v_3, e_1\rangle = \left\langle x^2, \frac{1}{2}\right\rangle = \frac{1}{2}(1+0+1+4) = 3,$$

and

$$\begin{aligned}\langle v_3, e_2\rangle &= \left\langle x^2, \frac{x-\frac{1}{2}}{\sqrt{5}}\right\rangle \\ &= \frac{1}{\sqrt{5}}\left(\left(-\frac{3}{2}\right)\cdot 1 + \left(-\frac{1}{2}\right)\cdot 0 + \left(\frac{1}{2}\right)\cdot 1 + \left(\frac{3}{2}\right)\cdot 4\right) \\ &= \frac{5}{\sqrt{5}} = \sqrt{5}.\end{aligned}$$

Thus

$$w_3 = v_3 - \langle v_3, e_1\rangle e_1 - \langle v_3, e_2\rangle e_2 = x^2 - 3\cdot\frac{1}{2} - \sqrt{5}\cdot\frac{x-\frac{1}{2}}{\sqrt{5}} = x^2 - x - 1.$$

Compute the norm:

$$\|w_3\|^2 = 1^2 + (-1)^2 + (-1)^2 + 1^2 = 4, \qquad e_3 = \frac{w_3}{\|w_3\|} = \frac{x^2 - x - 1}{2}.$$

We have obtained the following orthonormal basis of W:

$$e_1 = \frac{1}{2}, \qquad e_2 = \frac{x-\frac{1}{2}}{\sqrt{5}}, \qquad e_3 = \frac{x^2-x-1}{2}.$$

3. Since $\dim V = 4$ and $\dim W = 3$, we have $\dim(W^\perp) = 1$. For $p \in W^\perp$, write its values at the four points as

$$A = p(-1), \quad B = p(0), \quad C = p(1), \quad D = p(2).$$

The condition $p \perp W$ is equivalent to $\langle p, q\rangle = 0$ for each $q \in \{1, x, x^2\}$; that is,

$$A+B+C+D = 0, \qquad (-1)A + 0\cdot B + 1\cdot C + 2D = 0, \qquad 1\cdot A + 0\cdot B + 1\cdot C + 4D = 0.$$

Solving the system gives a vector of values proportional to $(-1, 3, -3, 1)$, so we may

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EXERCISE 13.5 CONTINUED

choose

$$p(-1) = -1, \quad p(0) = 3, \quad p(1) = -3, \quad p(2) = 1.$$

Find a polynomial of degree at most 3 with these values: write $p(x) = ax^3+bx^2+cx+d$. From $p(0) = 3$, we obtain $d = 3$, and also

$$\begin{aligned} p(1) = -3 &\implies a + b + c = -6, \\ p(-1) = -1 &\implies -a + b - c = -4, \\ p(2) = 1 &\implies 8a + 4b + 2c = -2. \end{aligned}$$

The first two equations give $b = -5$, hence $a+c = -1$. The third gives $4a+2b+c = -1$, so $4a - 10 + c = -1$, or $4a + c = 9$. Subtracting yields $3a = 10$, so

$$a = \frac{10}{3}, \qquad c = -\frac{13}{3}, \qquad d = 3, \qquad b = -5.$$

Multiply by 3 to avoid fractions and obtain a spanning vector:

$$u(x) := 10x^3 - 15x^2 - 13x + 9.$$

Hence

$$W^{\perp} = \operatorname{span}\{u\}.$$

For a unit vector, note that $\|u\|^2 = |-3|^2 + |9|^2 + |-9|^2 + |3|^2 = 180$, so

$$\frac{u}{6\sqrt{5}}$$

is an orthonormal basis of $W^{\perp}$.

EXERCISE 13.6

Continue with $V = \mathbb{C}_3[x]$ and the inner product

$$\langle f, g \rangle = f(-1)\,\overline{g(-1)} + f(0)\,\overline{g(0)} + f(1)\,\overline{g(1)} + f(2)\,\overline{g(2)},$$

and take $W = \mathbb{C}_2[x]$.
Write $P_W : V \to W$ for the orthogonal projection onto W.

1. Prove that P_W is linear and that $P_W^2 = P_W$.
2. Compute $P_W(x^3)$ explicitly using the orthonormal basis of W found in the preceding exercise.
3. Define $T : V \to V$ by
$$(Tf)(x) := f(x+1) - f(x).$$
Find all $q \in W$ such that $T(q) \in W^{\perp}$. Is the solution unique? Also find the unique solution satisfying $\langle q, 1 \rangle = 1$, and determine whether $\|q\| = 1$ would guarantee uniqueness.

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EXERCISE 13.6 CONTINUED

WORKED SOLUTION

Use the orthonormal basis of W:

$$e_1 = \frac{1}{2}, \qquad e_2 = \frac{x - \frac{1}{2}}{\sqrt{5}}, \qquad e_3 = \frac{x^2 - x - 1}{2}.$$

1. Every $f \in V$ has a unique orthogonal decomposition $f = w + u$, where $w \in W$ and $u \in W^\perp$. Define $P_W(f) := w$. Linearity: if $f = w + u$ and $g = w' + u'$ with $w, w' \in W$ and $u, u' \in W^\perp$, then

$$\alpha f + \beta g = (\alpha w + \beta w') + (\alpha u + \beta u')$$

where $\alpha w + \beta w' \in W$ and $\alpha u + \beta u' \in W^\perp$, so

$$P_W(\alpha f + \beta g) = \alpha w + \beta w' = \alpha P_W(f) + \beta P_W(g).$$

Furthermore, if $f = w + u$, then $P_W(f) = w \in W$, so $P_W(P_W(f)) = P_W(w) = w$; hence $P_W^2 = P_W$.

2. Since e_1, e_2, e_3 are orthonormal, for every $f \in V$,

$$P_W(f) = \sum_{i=1}^{3} \langle f, e_i \rangle e_i.$$

Compute this for $f(x) = x^3$.

First,

$$\langle x^3, e_1 \rangle = \left\langle x^3, \frac{1}{2} \right\rangle = \frac{1}{2}\left((-1)^3 + 0^3 + 1^3 + 2^3\right) = \frac{1}{2}(-1 + 0 + 1 + 8) = 4.$$

Second,

$$\langle x^3, e_2 \rangle = \left\langle x^3, \frac{x - \frac{1}{2}}{\sqrt{5}} \right\rangle = \frac{1}{\sqrt{5}} \sum_{t \in \{-1,0,1,2\}} \left(t - \tfrac{1}{2}\right) t^3.$$

Compute the sum:

$$\begin{aligned} t = -1: &\quad (-\tfrac{3}{2}) \cdot (-1) = \tfrac{3}{2}, \\ t = 0: &\quad (-\tfrac{1}{2}) \cdot 0 = 0, \\ t = 1: &\quad (\tfrac{1}{2}) \cdot 1 = \tfrac{1}{2}, \\ t = 2: &\quad (\tfrac{3}{2}) \cdot 8 = 12. \end{aligned}$$

Thus the sum is 14, giving

$$\langle x^3, e_2 \rangle = \frac{14}{\sqrt{5}}.$$

Third,

$$\langle x^3, e_3 \rangle = \left\langle x^3, \frac{x^2 - x - 1}{2} \right\rangle = \frac{1}{2} \sum_t (t^2 - t - 1)\, t^3.$$

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EXERCISE 13.6 CONTINUED

Compute:

$$\begin{aligned} t=-1: &\quad (1+1-1)\cdot(-1)=-1,\\ t=0: &\quad (-1)\cdot 0=0,\\ t=1: &\quad (1-1-1)\cdot 1=-1,\\ t=2: &\quad (4-2-1)\cdot 8=8. \end{aligned}$$

Thus the sum is 6, giving

$$\langle x^3, e_3\rangle = 3.$$

Consequently,

$$P_W(x^3) = 4e_1 + \frac{14}{\sqrt{5}}e_2 + 3e_3 = 4\cdot\frac{1}{2} + \frac{14}{\sqrt{5}}\cdot\frac{x-\frac{1}{2}}{\sqrt{5}} + 3\cdot\frac{x^2-x-1}{2}.$$

that is,

$$P_W(x^3) = 2 + \frac{14}{5}\left(x-\frac{1}{2}\right) + \frac{3}{2}(x^2-x-1) = \frac{3}{2}x^2 + \frac{13}{10}x - \frac{9}{10}.$$

3. We found $W^\perp = \text{span}\{u\}$, where

$$u(x) = 10x^3 - 15x^2 - 13x + 9.$$

Seek $q \in W$ with $T(q) \in W^\perp$: that is, there is $\lambda \in \mathbb{C}$ such that

$$T(q) = \lambda u.$$

Write $q(x) = ax^2 + bx + c$.

Compute:

$$q(x+1) = a(x+1)^2 + b(x+1) + c = ax^2 + (2a+b)x + (a+b+c),$$

Therefore

$$(Tq)(x) = q(x+1) - q(x) = 2ax + (a+b).$$

Thus $T(q)$ is a polynomial of degree at most 1.

On the other hand, if $T(q) \in W^\perp$, then $T(q) \in W^\perp \cap W$ (because its degree is at most 1, so $T(q) \in W$). But $W \cap W^\perp = \{0\}$, so necessarily

$$T(q) = 0.$$

Hence

$$2ax + (a+b) \equiv 0 \quad \Rightarrow \quad 2a = 0,\ a+b=0 \quad \Rightarrow \quad a=0,\ b=0.$$

and therefore $q(x) = c$ is constant.

Conversely, $T(q) = 0 \in W^\perp$ for every constant, so every $q = c$ is a solution. Without an additional condition, the solution is therefore not unique. Under the condition $\langle q, 1\rangle = 1$, we obtain $4c = 1$, so the unique solution is $q = \frac{1}{4}$. In contrast, the condition $\|q\| = 1$ does not guarantee uniqueness. For $q = c$,

$$\|q\|^2 = \langle c, c\rangle = |c|^2 + |c|^2 + |c|^2 + |c|^2 = 4|c|^2,$$

so $\|q\| = 1 \Rightarrow |c| = \frac{1}{2}$. Over the reals there are two possibilities, $c = \pm\frac{1}{2}$; over the

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EXERCISE 13.6 CONTINUED

complex numbers there are infinitely many possibilities of modulus $\frac{1}{2}$.

EXERCISE 13.7 EXAM QUESTION

Let $V = \mathbb{R}_4[x]$, the space of real polynomials of degree at most 4, including the zero polynomial. Define $L : V \to \mathbb{R}^3$ by

$$L(p) = (p(0), p(1), p'(1)).$$

Equip V with the inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)\,dt + p(0)q(0) + p(1)q(1).$$

Write $W = \ker L$ and $s(x) = 1 + x + x^2 + x^3 + x^4$.

1. Prove that L is linear.
2. Compute $\ker L$ and $\operatorname{Im} L$, and find a basis of each.
3. Compute the orthogonal projection $P_W(s)$ for the given inner product.

WORKED SOLUTION

1. Let $p, q \in V$ and $a, b \in \mathbb{R}$. By the linearity of evaluation and differentiation,

$$\begin{aligned} L(ap + bq) &= ((ap + bq)(0), (ap + bq)(1), (ap + bq)'(1)) \\ &= (ap(0) + bq(0), ap(1) + bq(1), ap'(1) + bq'(1)) \\ &= aL(p) + bL(q). \end{aligned}$$

   so L is linear.

2. The condition $p \in \ker L$ is equivalent to the three conditions

$$p(0) = 0, \qquad p(1) = 0, \qquad p'(1) = 0.$$

   The first says that x divides p. The last two say that 1 is a root of multiplicity at least 2: if $p(x) = (x-1)q(x)$, then $p'(x) = q(x) + (x-1)q'(x)$, so $p'(1) = q(1)$. Thus $(x-1)^2$ divides p. Since x and $(x-1)^2$ have no common root, and p has degree at most 4,

$$W = \{x(x-1)^2(a + bx) : a, b \in \mathbb{R}\}.$$

   Define

$$w_1 = x(x-1)^2 = x^3 - 2x^2 + x, \qquad w_2 = x^2(x-1)^2 = x^4 - 2x^3 + x^2.$$

   The polynomials w_1, w_2 span W and are linearly independent, for example because they have different degrees. Hence (w_1, w_2) is a basis of W, and $\dim W = 2$.

   To determine the image, exhibit polynomials mapped to the three standard basis vectors:

$$h_1 = (x-1)^2, \qquad h_2 = 2x - x^2, \qquad h_3 = x^2 - x.$$

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EXERCISE 13.7 EXAM QUESTION CONTINUED

A direct calculation gives

$$L(h_1) = (1,0,0), \qquad L(h_2) = (0,1,0), \qquad L(h_3) = (0,0,1).$$

Thus $\operatorname{Im} L = \mathbb{R}^3$, with the standard basis as a basis of the image. As a dimension check, $\dim \ker L + \dim \operatorname{Im} L = 2 + 3 = 5 = \dim V$.

3. Seek $p = P_W(s)$ in the form $p = aw_1 + bw_2$. The definition of orthogonal projection requires $p \in W$ and $s - p \in W^\perp$. Since w_1, w_2 form a basis of W, it suffices to require

$$\langle s - p, w_1 \rangle = 0, \qquad \langle s - p, w_2 \rangle = 0.$$

Every $w \in W$ satisfies $w(0) = w(1) = 0$, so the endpoint terms in the inner product vanish whenever one factor is w_1 or w_2. First compute the inner products between the basis vectors:

$$\begin{aligned}
\langle w_1, w_1 \rangle &= \int_0^1 (t^2 - 4t^3 + 6t^4 - 4t^5 + t^6)\, dt \\
&= \frac{1}{3} - 1 + \frac{6}{5} - \frac{2}{3} + \frac{1}{7} = \frac{1}{105}, \\
\langle w_1, w_2 \rangle &= \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7)\, dt \\
&= \frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{1}{280}, \\
\langle w_2, w_2 \rangle &= \int_0^1 (t^4 - 4t^5 + 6t^6 - 4t^7 + t^8)\, dt \\
&= \frac{1}{5} - \frac{2}{3} + \frac{6}{7} - \frac{1}{2} + \frac{1}{9} = \frac{1}{630}.
\end{aligned}$$

Multiplying out the polynomials also gives

$$\langle s, w_1 \rangle = \int_0^1 (t^7 - t^6 - t^2 + t)\, dt = \frac{1}{8} - \frac{1}{7} - \frac{1}{3} + \frac{1}{2} = \frac{25}{168},$$
$$\langle s, w_2 \rangle = \int_0^1 (t^8 - t^7 - t^3 + t^2)\, dt = \frac{1}{9} - \frac{1}{8} - \frac{1}{4} + \frac{1}{3} = \frac{5}{72}.$$

The orthogonality conditions are therefore

$$\begin{pmatrix} 1/105 & 1/280 \\ 1/280 & 1/630 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 25/168 \\ 5/72 \end{pmatrix}.$$

Multiplication by 2520 gives the system

$$24a + 9b = 375, \qquad 9a + 4b = 175.$$

Multiply the first equation by 4 and the second by 9, and subtract:

$$(96a + 36b) - (81a + 36b) = 1500 - 1575 \quad \Longrightarrow \quad 15a = -75.$$

Hence $a = -5$; the second equation gives $-45 + 4b = 175$, so $b = 55$. We conclude that

$$P_W(s) = -5x(x-1)^2 + 55x^2(x-1)^2$$

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EXERCISE 13.7 EXAM QUESTION CONTINUED

$$= 55x^4 - 115x^3 + 65x^2 - 5x.$$

The polynomial belongs to W by the factorization in the first line, and the two inner product equations ensure that its difference from s is orthogonal to all of W. These are exactly the two conditions that uniquely characterize the orthogonal projection.

13.3 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION 13.3

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w\rangle = \langle v, T^*w\rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^T$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

PROPOSITION 13.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram--Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

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EXERCISE 13.8

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \ldots, e_n)$ and real numbers $h_1, \ldots, h_n$ such that

$$He_j = h_j e_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_j e_j.$$

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \operatorname{diag}(h_1, \ldots, h_n), \qquad [K]_B = D_K := \operatorname{diag}(k_1, \ldots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H + K}{2}, \quad T^* = \frac{H - K}{2}.$$

Compute the two products separately, without assuming in advance that the factors commute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H + K)(H - K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

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EXERCISE 13.8 CONTINUED

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.
We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \dots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}}\, d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$\begin{aligned} AA^* &= (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*, \\ A^*A &= (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*. \end{aligned}$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

Additional practice

EXERCISE 13.9 FOUNDATION

In $\mathbb{R}^3$ with the standard inner product, consider the list

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (2, 2, 0)^T, \qquad v_3 = (1, 0, 1)^T.$$

Apply Gram--Schmidt, explicitly handling any zero residual. Find and check an orthonormal basis of the input span, and explain why the output does not contain three unit vectors.

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EXERCISE 13.9 FOUNDATION CONTINUED

WORKED SOLUTION

First, $\|v_1\| = \sqrt{2}$, so

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T.$$

For the second input,

$$\langle v_2, e_1 \rangle = 2\sqrt{2}, \qquad w_2 = v_2 - 2\sqrt{2}\, e_1 = 0.$$

We cannot normalize w_2; we skip it because $v_2 = 2v_1$. For the third input,

$$\langle v_3, e_1 \rangle = \frac{1}{\sqrt{2}}, \qquad w_3 = v_3 - \frac{1}{\sqrt{2}} e_1 = (1/2, -1/2, 1)^T.$$

We have $\|w_3\|^2 = 1/4 + 1/4 + 1 = 3/2$, so the second output vector is

$$e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

A direct check gives

$$\|e_1\|^2 = \frac{1+1}{2} = 1, \qquad \|e_2\|^2 = \frac{1+1+4}{6} = 1, \qquad \langle e_1, e_2 \rangle = \frac{1-1}{\sqrt{12}} = 0.$$

The vectors v_1, v_3 are linearly independent: in $av_1 + bv_3 = 0$, the third coordinate gives $b = 0$, and then $a = 0$. Thus the input span has dimension 2, and (e_1, e_2) is its orthonormal basis. It cannot contain three orthonormal vectors, since those vectors would be linearly independent.

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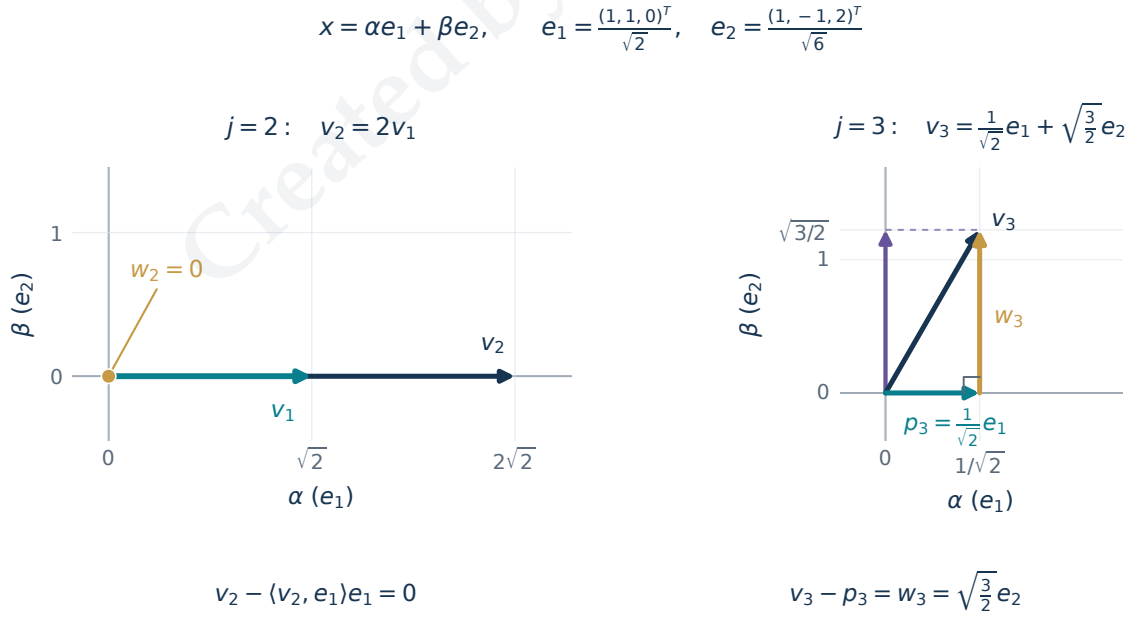


Figure 13.2. The axes α, β are coordinates in the orthonormal basis (e_1, e_2) of the plane $\text{span}\{v_1, v_3\}$: each point represents $x = \alpha e_1 + \beta e_2$. The input coordinates in this chart are $(\sqrt{2}, 0)^T$, $(2\sqrt{2}, 0)^T$, and $(1/\sqrt{2}, \sqrt{3/2})^T$, respectively. On the left, $v_2 = 2v_1$, so the residual $w_2 = 0$ must be skipped. On the right, subtract $p_3 = \langle v_3, e_1 \rangle e_1$ from v_3; the residual $w_3 = (1/2, -1/2, 1)^T = \sqrt{3/2}\, e_2$ is orthogonal to e_1.

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EXERCISE 13.10 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Prove that there is an orthogonal matrix Q for which both Q^TAQ and Q^TBQ are diagonal if and only if $AB = BA$.
2. Find such a matrix for

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 7 \end{pmatrix},$$

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^TAQ = D_A$ and $Q^TBQ = D_B$ are diagonal, then $A = QD_AQ^T$ and $B = QD_BQ^T$. Since diagonal matrices commute,

$$AB = QD_AD_BQ^T = QD_BD_AQ^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y\rangle = \langle x, By\rangle,$$

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = 2q_1, \quad Aq_2 = 2q_2, \quad Aq_3 = 5q_3, \qquad Bq_1 = 4q_1, \quad Bq_2 = 2q_2, \quad Bq_3 = 7q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^TAQ = \operatorname{diag}(2,2,5), \qquad Q^TBQ = \operatorname{diag}(4,2,7), \qquad Q^TABQ = \operatorname{diag}(8,4,35).$$

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EXERCISE 13.10 CHALLENGE CONTINUED

The choice within the two-dimensional eigenspace of A allows the basis to be adapted to B as well.

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